

Cloud Computing

Haystack Dense Retrieval Practice Document

Introduction

Cloud computing provides computing resources such as servers, storage, databases, networking, and software over a network. Organizations can use resources when needed instead of maintaining all infrastructure locally.

Service Models

The main cloud service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). IaaS provides virtualized infrastructure, PaaS provides application development platforms, and SaaS provides ready-to-use applications.

Deployment Models

Common deployment models include public cloud, private cloud, hybrid cloud, and community cloud. Public cloud resources are generally provided by a cloud provider, while hybrid environments combine different infrastructure types.

Benefits

Cloud computing can provide scalability, flexible resource usage, reduced infrastructure maintenance, rapid deployment, backup options, and access to services from different locations. Organizations can scale resources according to workload.

Challenges

Cloud environments require attention to security, privacy, access control, availability, compliance, cost management, and vendor dependency. Good architecture and monitoring help reduce operational risks.

This document is prepared as a small knowledge source for a Haystack BM25 versus dense retrieval experiment.